

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Basic Electronics Course Code: EC-111

Examination: B.Tech Scheme: CBS Branch - CSE (AI&ML) Semester: 2nd

Question Paper for 1st Periodical Examinations (March 2024)

Time: Two hours

Max. Marks: 50

Note: All questions are compulsory. Answer according to knowledge levels.

Course Outcomes: At the end of this course students will be able to:

CO112.1	K2	Understand the applications of diode in rectifiers, filter circuits and wave shaping.
CO112.2	K2	Understand the difference between Bipolar Junction Transistors and Field Effect Transistors and differentiate on basis of their characteristics.
CO112.3	K4	Design simple circuits like amplifiers (inverting and non-inverting), comparators, adders, integrator and differentiator using OPAMPS
CO112.4	K4	Compile the different building blocks in digital electronics using logic gates and implement simple logic function using basic universal gates

Knowledge levels :: K1=Remember, K2=Understand, K3=Apply, K4=Analyze, K5=Evaluate, K6=Create.

PART - A

Q.	(CO)	Question	K	Marks
Q1.	(CO112.1)	Define rectifier. Sketch a centre-tapped full-wave rectifier Bridge rectifier with construction and working. Show the appropriate waveforms. (i) Average Voltage (ii) Efficiency and (iii) Ripple factor	K2	(10)
Q2.	(CO112.1)	Derive the expressions for I_{dc} , V_{dc} , I_{rms} , V_{rms} , regulation, efficiency η , ripple factor and PIV of a full-wave rectifier (Both Center Tap and Bridge Rectifier). Show that the maximum efficiency of a full-wave rectifier is 81.2%.	K2	(10)
Q3.	(CO112.1)	Write a note on photodiode and phototransistor and mention their applications.	K1	(10)
Q4.	(CO112.2)	Explain the working of a Transistor as an amplifier? The collector and base current of an NPN transistor are measured as $I_c=5mA$, $I_b=50$ micro Ampere and $I_{cbo}=1$ micro Ampere. Determine β , α and I_e .	K1	(10)

PART - B

Q.5	(CO)	Answer in brief according to marks assigned.	K	(10)
a		Differentiate between drift current and diffusion current?		(2)
b		Explain the terms Knee voltage and breakdown voltage?		(2)
c		What is breakdown and what are its types in case of junction diode?		(2)
d		What is form factor and peak factor and what are its values of Half wave and Full wave rectifier?		(2)
e		What is quiescent point and explain load line concept?		(2)

(Course Advisor: Ankush Kapoor)

Vision: To achieve academic excellence and become pace setter by opting quality processes interwoven with freedom of flexibility and to mould professionals of superior quality dedicated to the progress and development of humanity.